SILVER OAK UNIVERSITY
EDUCATION TO INNOVATION

School of Technology, Design and Computer Application

College Of Technology

Bachelor of Technology (Information Technology)

Semester:	V	**Academic Year:**	2026-27
Course Name:	Data Science	**Course Code:**	2010043361

Question bank

Sr. No.	Question Text	Marks	CO
Unit No 1: Introduction to Data Science			
1	Define Data Science and explain its importance.	4	CO1
2	Differentiate between AI, ML, and Data Science.	4	CO1
3	List the types of data with examples.	4	CO1
4	Explain the Data Science lifecycle.	6	CO1
5	Describe the CRISP-DM process in Data Science.	6	CO1
6	Explain various data sources used in Data Science projects.	6	CO1
7	Discuss ethical issues in Data Science and their implications.	8	CO1
8	Compare structured, semi-structured, and unstructured data with suitable examples.	8	CO1
9	Explain the evolution of Data Science from traditional statistics to modern AI-driven systems.	8	CO1
Unit No 2: Data Analytics			
10	What is NumPy? Mention its key features.	4	CO2
11	State the purpose of Pandas in Data Analysis.	4	CO2
12	List the major visualization libraries available in Python.	4	CO2
13	Explain the architecture of the Python Data Science ecosystem.	6	CO2
14	Describe the role of NumPy and Pandas in Data Analytics.	6	CO2
15	Explain the applications of Data Science in healthcare and finance.	6	CO2
16	Compare Matplotlib and Seaborn libraries.	8	CO2
17	Analyze the use of Data Science in business decision-making.	8	CO2
18	Explain how Python libraries facilitate end-to-end data analysis.	8	CO2

Unit No 3: Data Preprocessing and Exploratory Data Analysis (EDA)			
19	What are missing values? Mention two methods to handle them.	4	CO3
20	Define feature engineering.	4	CO3
21	What is data normalization?	4	CO3
22	Explain different techniques for data cleaning.	6	CO3
23	Discuss data transformation and normalization methods.	6	CO3
24	Explain the role of EDA in Data Science projects.	6	CO3
25	Apply EDA techniques to identify patterns and outliers in a dataset.	8	CO3
26	Analyze the significance of feature engineering in machine learning models.	8	CO3
27	Explain various data visualization techniques used in EDA with suitable examples.	8	CO3
Unit No 4: Machine Learning for Data Science			
28	Define supervised learning and unsupervised learning.	4	CO4
29	What is overfitting?	4	CO4
30	Define train-test split.	4	CO4
31	Explain the workflow of a Machine Learning project.	6	CO4
32	Describe various model evaluation metrics for classification problems.	6	CO4
33	Explain the causes and remedies of overfitting and underfitting.	6	CO4
34	Compare supervised and unsupervised learning algorithms with examples.	8	CO4
35	Analyze the performance of a machine learning model using evaluation metrics.	8	CO4
36	Explain the complete process of developing a machine learning model from data preparation to evaluation.	8	CO4
Unit No 5: Deep Learning			
37	Define Artificial Neural Network (ANN).	4	CO4
38	What is an activation function?	4	CO4
39	List different types of activation functions.	4	CO4
40	Explain the structure of a biological neuron and an artificial neuron.	6	CO4
41	Describe the working of a feed-forward neural network.	6	CO4

42	Explain the training and testing process of a neural network.	6	CO4
43	Compare Sigmoid, Tanh, and ReLU activation functions.	8	CO4
44	Analyze the impact of activation functions on neural network performance.	8	CO4
45	Explain the architecture and working of an Artificial Neural Network with a neat diagram.	8	CO4
Unit No 6: Introduction to Natural Language Processing (NLP)			
46	Define Natural Language Processing (NLP).	4	CO5
47	List any four applications of NLP.	4	CO5
48	What is text representation?	4	CO5
49	Explain the scope and challenges of NLP.	6	CO5
50	Describe the basic NLP pipeline.	6	CO5
51	Explain various text representation techniques.	6	CO5
52	Analyze the role of NLP in modern AI systems.	8	CO5
53	Compare Bag-of-Words and TF-IDF text representation techniques.	8	CO5
54	Explain real-world applications of NLP in chatbots, sentiment analysis, and machine translation.	8	CO5

Course Coordinator **Head of Department**